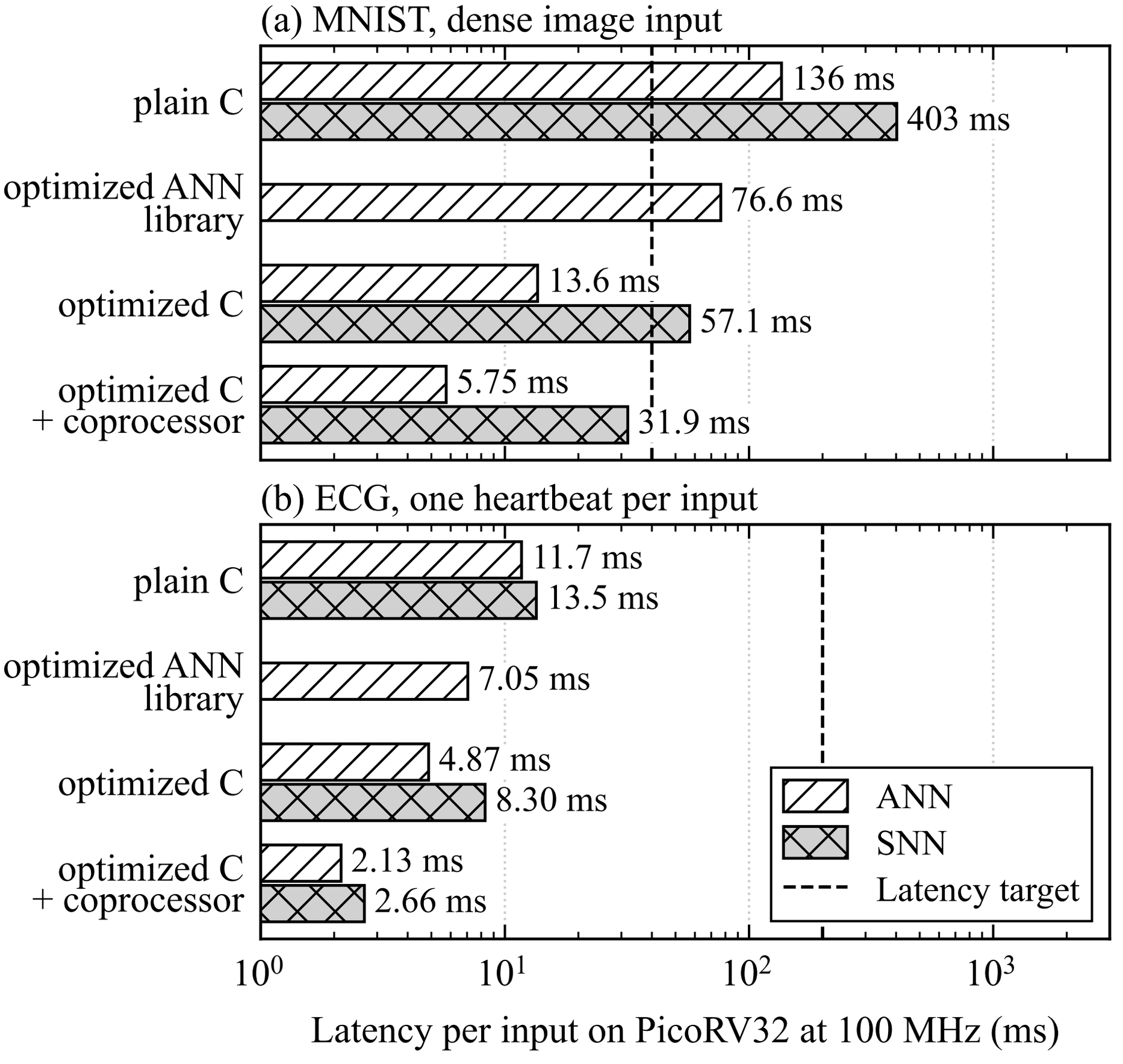
(a) MNIST, dense image input
plain C
136 ms
403 ms
optimized ANN library
76.6 ms
optimized C
13.6 ms
57.1 ms
optimized C + coprocessor
5.75 ms
31.9 ms
(b) ECG, one heartbeat per input
plain C
11.7 ms
13.5 ms
optimized ANN library
7.05 ms
optimized C
4.87 ms
8.30 ms
optimized C + coprocessor
2.13 ms
2.66 ms
ANN
SNN
Latency target
10^0
10^1
10^2
10^3
Latency per input on PicoRV32 at 100 MHz (ms)